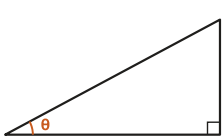
Trigonometry: from fingerprints to proofs

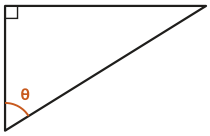
1 · Fingerprints NO CALCULATOR

Every angle has its own fixed ratios. Show your working, and write a sentence when the question says **explain**.

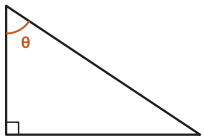
1 In each triangle, write **opp**, **adj** and **hyp** next to the correct sides, for the angle marked θ. [3 marks]



(a)



(b)

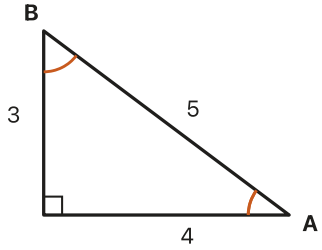


(c)

2 **(a)** Write sin A, cos A and tan A as fractions. [3]

(b) Write sin B and cos B as fractions. [2]

(c) Compare your answers to (a) and (b). What do you notice? **Explain** why this must happen in every right-angled triangle. [2]

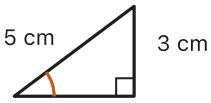


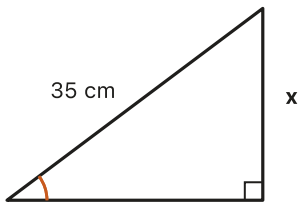
Not drawn accurately

3 These two triangles have the same angles. In the small one, the opposite side is 3 cm and the hypotenuse is 5 cm.

(a) Without a calculator, work out the side marked **x** in the big triangle. [2]

(b) **Explain** in one sentence, using *because*, why you did not need to know the angle. [1]





Not drawn accurately

4 CALCULATOR Your calculator says sin 35° = 0.574 (to 3 decimal places).

(a) Finish the sentence: "In *every* right-angled triangle with a 35° angle, …" [1]

(b) A triangle has a 35° angle and a hypotenuse of 20 cm. Work out the side opposite the 35°. [2]

(c) Another has a 35° angle and the side opposite it is 8.61 cm. Work out its hypotenuse. [2]

(b) ..... cm

(c) ..... cm

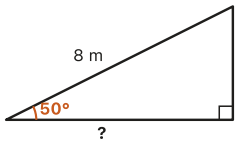
5 **Think.** (a) Can the sine of an angle in a right-angled triangle ever be bigger than 1? **Explain** using the sides of the triangle. (b) Can the tangent be bigger than 1? If so, give an example angle. (c) Without a calculator, find the angle whose tangent is exactly 1, and **explain** how you know. [4 marks]

Check: does each answer make sense with the picture?

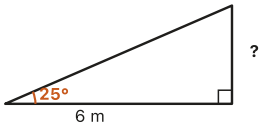
2 · Sides and angles CALCULATOR

Method, every time: ① label opp, adj, hyp from the angle ② circle the two sides you know or want ③ pick the ratio that uses exactly those two ④ write the equation, then solve ⑤ **check**: is the hypotenuse still the longest side? Give answers to 1 decimal place.

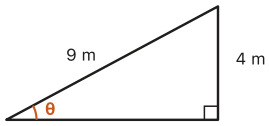
6 Don't work anything out yet. For each triangle, circle the ratio you would use: **sin cos tan**. [3 marks]



(a) sin cos tan

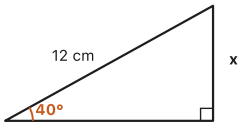


(b) sin cos tan

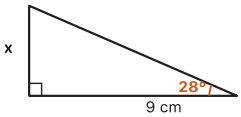


(c) sin cos tan

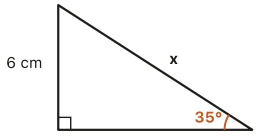
7 Work out the length **x** in each triangle. Part (c) is harder: **x** ends up on the bottom of the fraction. [7 marks]



(a) [2]



(b) [2]



(c) [3]

Not drawn accurately

x = ...

x = ...

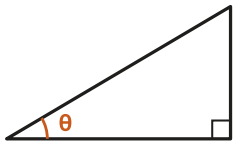
x = ...

8 Now the angle is the unknown. Use the inverse button ($\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$). Work out θ in each triangle. [4 marks]

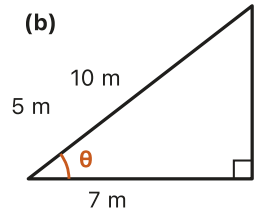
(a) θ = ...°

(b) θ = ...°

(a)



(b)



9 Back to the tree. You stand 24 m from the bottom of a tree on flat ground. Your eyes are 1.5 m above the ground. You measure the angle up to the top of the tree as 32°. **Draw a labelled diagram**, then work out the height of the tree. [4 marks]

height = ... m

10 **Spot the mistake.** A 5 m ladder leans against a wall. It makes an angle of 70° with the ground. Sam works out how high up the wall it reaches: [3 marks]

```
height = 5 × cos 70° = 1.71 m
```

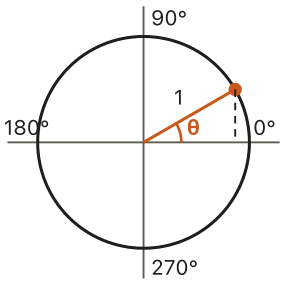
(a) Without doing any calculation, how can you tell that 1.71 m must be wrong? (b) What mistake did Sam make? (c) Work out the correct height.

3 · Beyond the triangle NO CALCULATOR

Remember the circle of radius 1: the point at angle θ is **cos θ across** and **sin θ up**. Angles go anticlockwise from the right.

11 Use the circle only, no calculator. Fill in the table, then for **one** of the boxes **explain** how the circle tells you the answer. [4 marks]

	0°	90°	180°	270°	360°
sin					
cos					



12 sin 30° = 0.5. [5 marks]

(a) Find another angle between 0° and 360° whose sine is also 0.5. **Explain** using the circle (sketch it). [2]

(b) Is sin 200° positive or negative? **Explain** how you know. [1]

(c) Write down an angle whose cosine is negative, and one whose sine and cosine are *both* negative. [2]

13 In Room 5 you showed that **$(\sin\theta)^2 + (\cos\theta)^2 = 1$** for every angle. [5 marks]

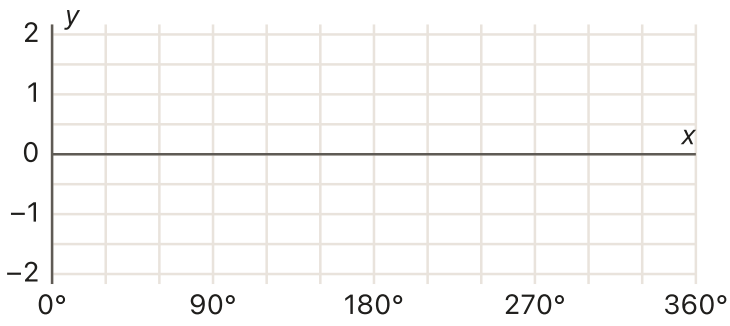
(a) sin θ = 0.8. Find **two** possible values of cos θ. Why are there two? [2]

(b) Angle A is in a right-angled triangle and cos A = $^5/_{13}$. Without a calculator, find sin A and tan A as fractions. (Hint: sketch a triangle that has this cosine.) [3]

14 (a) Sketch **y = sin x** for 0° to 360° on the grid. Mark where it crosses the x-axis. [3 marks]

(b) In a different colour, sketch **y = 1 + sin x**.

(c) In words: what did the "1 +" do to the graph?



15 A Ferris wheel seat starts level with the centre and turns anticlockwise. After the wheel turns through θ, its height in metres is **height = 20 + 15 sin θ**. [6 marks]

(a) How high is the centre of the wheel? What is its radius? [1]

(b) What are the highest and lowest heights of the seat? **Explain** using the largest and smallest values sin can take. [2]

(c) Use your answer to 12(a) and the circle to find the height after 210°. [1]

(d) The seat is 27.5 m high twice on the way round. At which two angles? [2]

Challenge: think it through

4 · Exact values, real exam questions, two triangles

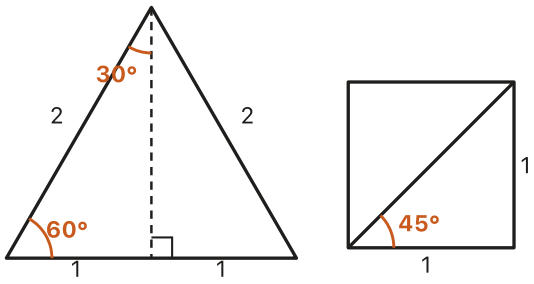
> $\sqrt{3}$ is the number that squares to 3, so $\sqrt{3} \times \sqrt{3} = 3$. Leave $\sqrt{3}$ as it is: it is *exact*, and 1.732… is not.

16 NO CALCULATOR [6 marks]

(a) An equilateral triangle with sides of 2 is cut in half (left). Use Pythagoras to **show** that the dashed height is $\sqrt{3}$. [2]

(b) Use the half-triangle to write **exact** values for sin, cos and tan of 30° and of 60°. [3]

(c) A square with sides of 1 is cut along a diagonal (right). Find tan 45° and sin 45°. [1]



REAL GCSE QUESTION · AQA 8300/1H June 2018, Higher Paper 1 (non-calculator), Q30 [3 marks]

17 Show that the value of cos 30° × tan 60° + sin 30° is an integer.

REAL GCSE QUESTION · AQA 8300/1H November 2022, Higher Paper 1 (non-calculator), Q23 [4 marks]

18 4 × sin 30° × tan 30° × cos 30° = sin *y*

Work out one possible value of *y*.
You must show your working.

y = ……°

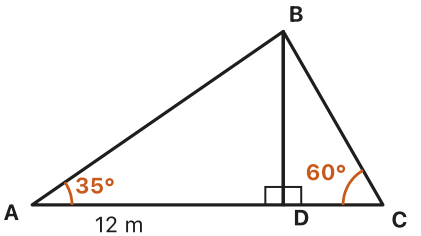
Second way, if you finish: tan 30° = sin 30° ÷ cos 30° (why?). Put that in and watch what cancels.

19 CALCULATOR A vertical pole BD stands on flat ground. [6 marks]
From A, 12 m from the foot of the pole, the angle up to the top B is 35°. A wire runs from B down to C, making 60° with the ground.

(a) Work out the height of the pole BD. [2]

(b) Work out the length of the wire BC. [2]

(c) Sam rounded BD to 8 m before doing (b). What answer did he get? **Explain** why you should keep the full number in your calculator. [2]



Not drawn accurately

20 Lily says: *"sin 40° = 0.643, so a triangle with a 40° angle has an opposite side of 0.643 cm."* Is she right? **Explain**, using what a ratio means. [2 marks]

Total: 87 marks · Questions 17 and 18 are real GCSE Higher questions